dirac16complex as dark matter and dark energy

An analysis based on the accurate approximate solutions of the coupled field equations of the 7+1 model in the three gravity regimes — section 24 of the notebook 7+1.nb

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Abstract

Can the 16-component field dirac16complex of the 7+1 model be the dark matter of the universe, and can it be the dark energy? This document answers on the basis of the accurate approximate solutions of the coupled field equations in the three gravity regimes of the model (primordial, modern, far future), the couplings of the model, and present observational data. Every statement is marked DERIVED (an exact consequence of the field equations, checked), ESTIMATE (an order-of-magnitude statement with stated assumptions) or OPEN (not decided by the model). The verdict: dirac16complex has every property the solutions and the observations demand of dark matter — it is cold, dark and collisionless and enters the late-universe solution exactly as the cold-dark-matter density — provided the auxiliary space is stabilized and its Yukawa portal is very weak; but the minimal model does not produce the observed abundance with couplings of natural size (OPEN). It cannot be the dark energy: as matter its equation of state never falls below $w = -0.06$ in a stable state, and its vacuum energy is a constant contribution to the cosmological constant ($w = -1$ exactly). Section 24 of 7+1.nb contains 20 checks (20 True); the independent Python program tools/ref_dark.py contains 20 (20 True); the 29 groups of numbers printed by both agree to a largest relative difference of 1.4×10^{-7}.

Contents

1. Results at a glance 2

2. The model, the conventions and the objects used 2

3. What the observations require (section 24.1) 4

4. dirac16complex as cold matter in the solutions of the three regimes (section 24.2, DERIVED) 5
4.1 The stress tensor of a massive particle . . . 5
4.2 Regime I (primordial): the gas wins, and needs the stabilizer . . . 5
4.3 Regime II (modern): the dark matter of the late-universe solution . . . 5
4.4 Regime III (far future): the dark matter dilutes away . . . 6

5. The lightest mode: dark, collisionless, and stable or not (section 24.3, ESTIMATE) 6

6. The abundance (section 24.4, ESTIMATE; the outcome is OPEN) 7
6.1 From the solutions of regimes I and II: overproduction . . . 7
6.2 After the dilution: the portals . . . 7
6.3 Conclusion on the abundance . . . 9

7. dirac16complex and dark energy (section 24.5) 9
7.1 Matter cannot be dark energy (DERIVED) . . . 9
7.2 A condensate is only a cosmological constant (DERIVED) . . . 10
7.3 Vacuum energy on the auxiliary torus (DERIVED, with an ESTIMATE of its size) . . . 10
7.4 The prediction $w = -1$, and DESI . . . 10

8. Verdict (section 24.6) 11

9. Validity and limitations 11

10. Verification 12

11. How to reproduce 14

1. Results at a glance

question	answer	status
Is dirac16complex matter cold, and can it be the Ω_c of the late-universe solution?	yes: the lightest mode has $p_a/\varepsilon = k_a^2/(3(m^2a^2+k_a^2)) \to 0$, $\varepsilon \propto a^{-3}$; it is exactly the term $\Omega_c a^{-3}$ (section 4)	DERIVED
Does it disturb the auxiliary space today?	no: it displaces the radion by $\delta\beta = H_0^2\Omega_c/(2m_r^2) = 2.7\times10^{-87}$ (section 4.3)	DERIVED
Does it need the stabilizing potential?	yes: without it a bulk gas re-expands the auxiliary space, $b \propto \tau^{2/5}$ (section 4.2)	DERIVED
Is it dark?	$\sigma_{\rm SI} = 9.6\times10^{-104}$ cm^2 at $m = 40$ GeV for a Higgs portal of natural size (LZ: 2.2×10^{-48} cm^2) (section 5)	ESTIMATE
Is it collisionless?	$\sigma/m = 4.9\times10^{-79}$ cm^2/g at $m = 1$ GeV (Bullet Cluster: < 0.7 cm^2/g) (section 5)	ESTIMATE
Is it stable?	absolutely without the Yukawa portal; with it, longer than the age of the universe only for $m < 40.2$ MeV$\times c_y^{-2/3}$ (section 5)	ESTIMATE
Does the model make the observed abundance $mY = 4.37\times10^{-10}$ GeV?	its primordial gas and radion make far too much (overabundance $1.5\times10^7\, T_{\rm RH}$/GeV, brane share 7.6×10^{-14}); after the dilution its moduli problem requires, its portals with natural couplings make far too little (the Higgs portal would need $c_P = 149$) (section 6)	OPEN
Can its matter be the dark energy?	no: $w = p/\varepsilon \geq -0.0580$, and $p < 0$ only where uniform matter is unstable or metastable (section 7.1)	DERIVED
Can its vacuum energy be the dark energy?	only as a constant contribution to Λ_4 ($w = -1$); its Casimir energy on the auxiliary torus is $6.8\times10^{34}\,\rho_\Lambda$ if massless (section 7.3)	DERIVED / ESTIMATE
Evolving dark energy (DESI DR2)?	the model predicts $w = -1$ exactly; a confirmed $w_0 > -1$, $w_a < 0$ would contradict it (section 7.4)	DERIVED

2. The model, the conventions and the objects used

Units $\hbar = c = 1$; coordinates $X = (x_0, x_1, \ldots, x_7)$: x_0 is time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ the four auxiliary space-like coordinates; $y = (x_4, \ldots, x_7)$. The frame metric is $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$, simply the squares of the octonion basis.

The signature: $\{7, 1, 0\}$ and $\mathrm{diag}(+1, -1, \ldots, -1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1, +1, \ldots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like $= +1$, time-like $= -1$). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \ldots, 7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\,\eta_{AB}\,1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p, q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1, 7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det\eta = \det\eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7, 1, 0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB}x^A x^B$.

The field. dirac16complex is a 16-component complex spinor field $\psi(X)$ (anticommuting operators when quantized) with the real 16×16 Dirac matrices Γ_A, $\Gamma_A\Gamma_B+\Gamma_B\Gamma_A=2\eta_{AB}1_{16}$, $\Gamma^A=\eta^{AA}\Gamma_A$, the Dirac conjugate $\bar\psi=\psi^\dagger\Gamma_0$, the frame matrices $\Gamma^\mu=e_A{}^\mu\Gamma^A$ ($g_{\mu\nu}=\eta_{AB}e^A{}_\mu e^B{}_\nu$) and the spin connection $\Omega_\mu=\frac14\eta_{AC}\omega_\mu{}^C{}_B\Gamma^A\Gamma^B$. **The action of the model** (gravity with the Lovelock terms $k=0,\dots,3$, $\kappa=8\pi G_8=M_8^{-6}$; dirac16complex with its mass and nine self-couplings; a stabilizing potential of the auxiliary space; on the world brane $W=\{y=0\}$ with induced metric h: a tension, the Standard Model and three portals):

$$\begin{aligned}S=&-\frac{1}{2\kappa}\int d^8X\sqrt{|g|}\big(2\Lambda+R+\alpha_2L_2+\alpha_3L_3\big)+\int d^8X\sqrt{|g|}\Big[\bar\psi\,i\Gamma^\mu(\partial_\mu+\Omega_\mu)\psi-m\bar\psi\psi\\&-\frac12\sum_{r=0}^{8}g_r\sum_{|K|=r}\eta_K(\bar\psi\Theta^K\psi)^2-U(b)\Big]\\&+\int_W d^4x\sqrt{|h|}\Big[-f_W+\mathcal{L}_{\rm SM}-\sum_{i,s}y_{is}\bar L_i\tilde\Phi P_R\psi_s+\text{h.c.}-G_P^f(\bar\psi\Gamma^\mu\psi)(\bar f\gamma_\mu f)-\lambda_P\Phi^\dagger\Phi\,\bar\psi\psi\Big],\\L_2=&R^2-4R_{\mu\nu}R^{\mu\nu}+R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma},\qquad L_3=\frac18\delta^{\alpha_1\beta_1\alpha_2\beta_2\alpha_3\beta_3}_{\gamma_1\delta_1\gamma_2\delta_2\gamma_3\delta_3}R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}R^{\gamma_2\delta_2}{}_{\alpha_2\beta_2}R^{\gamma_3\delta_3}{}_{\alpha_3\beta_3},\\U(b)=&U_0+\frac{K_U}{2}\ln^2\frac{b}{b_0},\qquad b^8=\det(-g_{kl})\ (k,l=4,\dots,7),\end{aligned}$$

where $\Theta^K=i^{\sigma_r}\Gamma^{A_1}\cdots\Gamma^{A_r}$ for $K=(A_1<\cdots<A_r)$, $\sigma_r=\frac{r(r-1)}{2}\bmod 2$, $\eta_K=\eta_{A_1A_1}\cdots\eta_{A_rA_r}$; $\delta^{\cdots}_{\cdots}$ is the generalized Kronecker delta (the determinant of Kronecker deltas); $\mathcal{L}_{\rm SM}$ is the Standard Model Lagrangian in the field of h (Higgs doublet Φ, $\tilde\Phi=i\sigma_2\Phi^*$, lepton doublets L_i, fermions f); ψ_s are brane-right-handed parts of ψ at the brane; y_{is}, G_P^f, λ_P are the Yukawa (neutrino), vector and Higgs portals.

The homogeneous metric of the three regimes and its Hubble rates (a prime is d/dx_0; τ is the time since the singularity):

$$ds^2=dx_0^2-a^2(dx_1^2+dx_2^2+dx_3^2)-b^2(dx_4^2+\cdots+dx_7^2),\qquad H_a=\frac{a'}{a},\qquad H_b=\frac{b'}{b},\qquad \beta=\ln\frac{b}{b_0}.$$

Every source is $T^\mu{}_\nu=\mathrm{diag}(\varepsilon,-p_a,-p_a,-p_a,-p_b,-p_b,-p_b,-p_b)$; for matter whose comoving energy (at fixed occupation numbers) is $W(a,b)$,

$$\varepsilon=\frac{W}{a^3b^4},\qquad p_a=-\frac{\partial_aW}{3a^2b^4},\qquad p_b=-\frac{\partial_bW}{4a^3b^3},\qquad \varepsilon'+3H_a(\varepsilon+p_a)+4H_b(\varepsilon+p_b)=0,$$

and the field equations $\sum_k\alpha_kE^{(k)\mu}{}_\nu=\kappa T^\mu{}_\nu$ ($\alpha_0=2\Lambda$, $\alpha_1=1$) become three equations, $\sum_k\alpha_kE^{(k)0}{}_0=\kappa\varepsilon$, $\sum_k\alpha_kE^{(k)1}{}_1=-\kappa p_a$, $\sum_k\alpha_kE^{(k)4}{}_4=-\kappa p_b$; the first is the constraint

$$3(H_a^2+4H_aH_b+2H_b^2)-12\alpha_2H_b(4H_a^3+18H_a^2H_b+12H_aH_b^2+H_b^3)+360\alpha_3H_a^2H_b^3(4H_a+3H_b)-\Lambda=\kappa\varepsilon.$$

The radion. In Einstein gravity the 11 and 44 equations and the constraint give exactly

$$\beta''+3H_a\beta'+4\beta'^2=\kappa\Big(\frac{\varepsilon}{6}-\frac{p_a}{2}+\frac{p_b}{3}\Big)+\frac{\Lambda}{3},$$

so the potential (which contributes $\varepsilon=U$, $p_a=-U$, $p_b=-U-\frac b4U'$) gives the restoring term $-(\kappa K_U/12)\beta$: the radion mass is $m_r^2=\kappa K_U/12$. With the auxiliary space static the 00 and 11 equations are the Friedmann equations of the brane,

$$3H_a^2=8\pi G_4\rho_4+\Lambda_4,\qquad H_a'=-\frac{\kappa}{2}(\varepsilon+p_a),\qquad 8\pi G_4=\frac{\kappa}{V_\perp}=\frac{1}{M_{\rm Pl}^2},\qquad \Lambda_4=\Lambda+\kappa U_0+\frac{\kappa f_W}{V_\perp},$$

with $V_\perp=b_0^4L_c^4$ the proper volume of the auxiliary torus (L_c its coordinate period), $\rho_4=V_\perp\varepsilon$ the energy per brane volume and $M_{\rm Pl}$ the reduced Planck mass. The canonical radion is $\phi=\sqrt{12}M_{\rm Pl}\beta$; it couples to brane fields through $(\phi/(\sqrt3M_{\rm Pl}))T^\mu{}_\mu$.

Kaluza–Klein modes. On the torus a field is a sum of modes $e^{i\vec q\cdot\vec y}$ with $\vec q=2\pi\vec n/L_c$, $\vec n\in\mathbb{Z}^4$; at $b=b_0$ their four-dimensional masses are

$$M_n=\sqrt{m^2+\mu_n^2},\qquad \mu_n=\frac{2\pi|\vec n|}{\ell_\perp},\qquad \ell_\perp=V_\perp^{1/4},\qquad \#\{\vec n:|\vec n|^2=k\}=r_4(k)=8\sigma(k)-32\sigma(k/4),$$

(σ the divisor sum, $\sigma(k/4) = 0$ unless $4 \mid k$: Jacobi's four-square theorem). Each level is four brane Dirac spinors (the 16 components of ψ split at the brane into four 4-component spinors). The number of levels with $\mu_n < \mu$ and their density are, in the continuum limit,

$$N(\mu) = \frac{\pi^2}{2}\Big(\frac{\mu\ell_\perp}{2\pi}\Big)^4 = \frac{V_\perp \mu^4}{32\pi^2}, \qquad dN = \frac{V_\perp \mu^3\, d\mu}{8\pi^2}.$$

The LIGHTEST MODE ($\vec{n} = 0$) is constant over the torus: $\psi(x, y) = \chi(x)/\sqrt{V_\perp}$ with canonically normalized four-dimensional fields χ (four Dirac spinors of mass m).

Physical parameters (as in the solutions of the modern regime; the mass m of dirac16complex is free):

quantity	value	meaning
M_8	10^4 GeV	reduced eight-dimensional Planck mass, $\kappa = M_8^{-6}$
$M_{\rm Pl}$	2.435×10^{18} GeV	reduced four-dimensional Planck mass, $M_{\rm Pl}^2 = V_\perp/\kappa = V_\perp M_8^6$
$V_\perp$, $\ell_\perp$	$M_{\rm Pl}^2/M_8^6 = 5.929 \times 10^{12}$ GeV^{-4}; 1560.45 GeV^{-1}	volume and size of the auxiliary torus
m_r	10 GeV	radion mass
α_2, α_3, f_W	$1/M_8^2$, $10^{-6}/M_8^4$, M_8^4	Gauss–Bonnet and cubic couplings, brane tension
m_h, v, G_F	125.20 GeV, $(\sqrt{2}G_F)^{-1/2} = 246.22$ GeV, 1.1663788×10^{-5} GeV^{-2}	Higgs mass, Higgs vacuum value, Fermi constant

3. What the observations require (section 24.1)

Data used, as reported by the collaborations (the confidence levels are theirs):

observation	result	source
cosmic microwave background	$\Omega_c h^2 = 0.120 \pm 0.001$, $\Omega_b h^2 = 0.0224 \pm 0.0001$, $H_0 = 67.4 \pm 0.5$ km/s/Mpc, $\Omega_m = 0.315 \pm 0.007$	Planck 2018, Astron. Astrophys. 641 (2020) A6
baryon acoustic oscillations	$w(a) = w_0 + w_a(1-a)$ with $w_0 > -1$, $w_a < 0$ preferred over Λ at 3.1σ (DESI + CMB), 2.8σ–4.2σ with supernovae	DESI DR2, arXiv:2503.14738
the same, with ACT	DESI + Planck PR4 + ACT DR6 + DES-Y5: $w_0 = -0.753 \pm 0.056$, $w_a = -0.84^{+0.23}_{-0.20}$ (4.2σ)	Garcia-Quintero et al., arXiv:2504.18464
direct detection	$\sigma_{\rm SI} < 2.2 \times 10^{-48}$ cm^2 at 40 GeV (90% CL)	LZ, 4.2 tonne-years, arXiv:2410.17036
self-interaction	$\sigma/m < 1.25$ cm^2/g (68% CL), < 0.7 cm^2/g with equal mass-to-light ratios	Randall et al., ApJ 679 (2008) 1173 (Bullet Cluster)
small-scale structure	thermal-relic warm dark matter heavier than 5.7 keV (95% CL)	Iršič et al., Phys. Rev. D 109 (2024) 043511 (Lyman-α)
the same, for decay-produced dark matter	warmer than a thermal relic of equal mass: about 7 keV where the thermal-relic bound is 3 keV	Ballesteros, Garcia, Pierre, JCAP 03 (2021) 101
Pauli principle	fermion with $g = 2$ internal states heavier than 0.13 keV (95% CL); the bound scales as $g^{-1/4}$	Alvey et al., MNRAS 501 (2021) 1188 (dwarf spheroidals)
X-ray lines	limits on the mixing of sterile neutrinos of 6–40 keV, from $\nu_s \to \nu\gamma$	NuSTAR, Roach et al., Phys. Rev. D 107 (2023) 023009
large extra dimensions	$M_D > 5.9$ to 11.2 TeV for $n = 6$ to 2 ($M_{\rm Pl}^2 = R^n M_D^{n+2}$, $V = (2\pi R)^n$)	Particle Data Group 2024 (ATLAS, 139 fb^{-1})

The required abundance. With the critical density per h^2, the entropy density today and its degrees of freedom,

$$mY = \frac{\Omega_c h^2\,(\rho_c/h^2)}{s_0}, \qquad \frac{\rho_c}{h^2} = 3\Big(100\,\frac{\rm km}{\rm s\,Mpc}\Big)^2 M_{\rm Pl}^2, \qquad s_0 = \frac{2\pi^2}{45} g_{*s0} T_{\rm CMB}^3, \qquad g_{*s0} = 2 + \frac{21}{4}\cdot\frac{4}{11},$$

($Y = n/s$ the comoving number density, $T_{\rm CMB} = 2.7255$ K) one finds $\Omega_c = 0.120/0.674^2 = 0.2642$, $\rho_c/h^2 = 1.0534 \times 10^{-5}$ GeV/cm^3, $s_0 = 2891.3$ cm^{-3} and

$$mY = 4.372 \times 10^{-10}\ \text{GeV}.$$

Collider bound. In the convention of the Particle Data Group, $V_\perp = (2\pi R)^4$ and $M_{\rm Pl}^2 = R^4 M_D^6$; with $M_{\rm Pl}^2 = V_\perp M_8^6$ this is $M_D = (2\pi)^{2/3} M_8 = 34.05$ TeV, above the strongest bound (11.2 TeV). **Pauli bound.** The lightest mode has $g = 16$ internal states (four brane Dirac spinors, each with two spin states, particle and antiparticle): $m > 0.13\,\text{keV} \times (2/16)^{1/4} = 0.0773$ keV.

4. dirac16complex as cold matter in the solutions of the three regimes (section 24.2, DERIVED)

4.1 The stress tensor of a massive particle

A particle of dirac16complex with comoving momenta k_a (ordinary space) and k_b (auxiliary space) has the comoving energy $\omega = W(a,b)$:

$$\omega = \sqrt{m^2 + \frac{k_a^2}{a^2} + \frac{k_b^2}{b^2}}, \qquad \varepsilon\, a^3 b^4 = \omega, \qquad p_a\, a^3 b^4 = \frac{k_a^2}{3a^2\omega}, \qquad p_b\, a^3 b^4 = \frac{k_b^2}{4b^2\omega},$$
$$\frac{p_a}{\varepsilon}\Big|_{k_b=0} = \frac{k_a^2}{3(m^2a^2+k_a^2)} \xrightarrow{k_a/a \ll m} 0, \qquad p_b\big|_{k_b=0} = 0, \qquad \varepsilon\big|_{k_a=k_b=0} = \frac{m}{a^3 b^4} \propto a^{-3} \quad (b \text{ fixed}).$$

The lightest mode ($k_b = 0$) is therefore cold, pressureless matter once $k_a/a \ll m$. Every other Kaluza–Klein level ($k_b/b_0 = \mu_n$) is matter of mass $M_n = (m^2 + \mu_n^2)^{1/2}$ with a static pressure along the auxiliary space, $p_b/\varepsilon = \mu_n^2/(4M_n^2) \le \frac{1}{4}$.

4.2 Regime I (primordial): the gas wins, and needs the stabilizer

Near $x_0 = 0$ any fluid with $p_a = w_a\varepsilon$ ($w_a < 1$) dominates the Lovelock terms. With b finite and $H_a = p/\tau$, the leading orders of the three equations are

$$\tau^3 \textstyle\sum \alpha_k E^{(k)0}{}_0 \to 48hp^3(30\alpha_3 h^2 - \alpha_2), \qquad \tau^3 \sum \alpha_k E^{(k)1}{}_1 \to 32hp^2(p-1)(18\alpha_3 h^2 - \alpha_2),$$
$$\tau^4 \textstyle\sum \alpha_k E^{(k)4}{}_4 \to p^3(p-1)(648\alpha_3 h^2 - 12\alpha_2), \qquad (H_b \to h),$$

so $h = h_* = -\sqrt{\alpha_2/(54\alpha_3)}$, $p = 1/(1+w_a)$ ($a \propto \tau^{3/4}$ for the gas, $w_a \to \frac{1}{3}$) and $\kappa\varepsilon\tau^3 \to \frac{64}{3}p^3\alpha_2|h_*|$. Later, in the Einstein regime, the gas fraction grows and the gas dominates; Einstein gravity with $p_a = w_a\varepsilon$, $p_b = w_b\varepsilon$ has the power laws $a \propto \tau^P$, $b \propto \tau^Q$ with

$$3(P^2 + 4PQ + 2Q^2) = \mathcal{C}, \qquad 3P^2 + 8PQ - 2P + 10Q^2 - 4Q = -w_a\mathcal{C},$$
$$3(2P^2 + 3PQ - P + 2Q^2 - Q) = -w_b\mathcal{C}, \qquad 3P(1+w_a) + 4Q(1+w_b) = 2;$$
$$(w_a, w_b) = (0, \tfrac{1}{4}) \;\Rightarrow\; (P,Q) = (0, \tfrac{2}{5}).$$

The gas of the coupled primordial solution has $p_b/\varepsilon \to \frac{1}{4}$ (its auxiliary momenta are blue-shifted) and reaches $\tau H_b = 0.3999953$ (exactly $\frac{2}{5}$ in the limit): ordinary space stops expanding and the AUXILIARY SPACE RE-EXPANDS. dirac16complex can therefore be dark matter only if the stabilizing potential $U(b)$ holds the auxiliary space.

4.3 Regime II (modern): the dark matter of the late-universe solution

With $U(b)$ the auxiliary space is caught when $H \approx m_r$; the radion then oscillates, $\beta \approx A\sin(m_r\tau + \vartheta)/(m_r(\tau - \tau_E))$ with $H_a + 2H_b \approx 2/(3(\tau - \tau_E))$, and the dirac16complex gas — its auxiliary momenta frozen into Kaluza–Klein masses — is four-dimensional matter: its share of the energy is constant, 0.00859579 and 0.00872803 at $\tau = 10^5$ and 3×10^5 (units $\ell_2 = \sqrt{\alpha_2} = 1$; for the normalization of the primordial gas). In the late universe the brane obeys

$$H_a = H_0\sqrt{\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda}, \qquad \Omega_m = \Omega_b + \Omega_c, \qquad \Omega_\Lambda = 1 - \Omega_m - \Omega_r,$$
$$t(a) \approx \frac{2}{3H_0\Omega_m^2}\Big[(\Omega_m a - 2\Omega_r)\sqrt{\Omega_r + \Omega_m a} + 2\Omega_r^{3/2}\Big] + \frac{2\,\text{arcsinh}\big(\sqrt{\Omega_\Lambda/\Omega_m}\,a^{3/2}\big)}{3H_0\sqrt{\Omega_\Lambda}} - \frac{2a^{3/2}}{3H_0\sqrt{\Omega_m}},$$

the composite solution (accurate to 2.2×10^{-4}; age 13.79 Gyr). Cold dirac16complex matter is exactly the term $\Omega_c a^{-3}$: dirac16complex CAN BE the cold dark matter of this solution without changing it. Being uniform over the torus, its energy density per brane volume is $\rho_{\rm DM}=V_\perp\varepsilon$ and it gravitates exactly like brane matter ($8\pi G_4=\kappa/V_\perp$). Through the radion equation (source $\kappa\varepsilon/6$ for $p_a=p_b=0$) it shifts the radion to

$$\delta\beta=\frac{\kappa\varepsilon}{6m_r^2}=\frac{\rho_{\rm DM}}{6M_{\rm Pl}^2m_r^2}=\frac{H_0^2\Omega_c}{2m_r^2}=2.730\times10^{-87}\qquad(\rho_{\rm DM}=3H_0^2M_{\rm Pl}^2\Omega_c,\ m_r=10\ \text{GeV}),$$

so $G_4\propto e^{-4\beta}$ does not change.

4.4 Regime III (far future): the dark matter dilutes away

For matter and Λ the brane solution is exact,

$$a(t)=\Big(\frac{\Omega_m}{\Omega_\Lambda}\Big)^{1/3}\sinh^{2/3}\Big(\frac{3}{2}H_\infty t\Big),\qquad H_\infty=H_0\Omega_\Lambda^{1/2},$$
$$\frac{\rho_{\rm DM}}{\rho_\Lambda}=\frac{\Omega_c a^{-3}}{\Omega_\Lambda}=\frac{\Omega_c}{\Omega_m}\,\frac{1}{\sinh^2(\frac{3}{2}H_\infty t)}\ \to\ 4\,\frac{\Omega_c}{\Omega_m}\,e^{-3H_\infty t},$$

the identity $\Omega_m a^{-3}/\Omega_\Lambda=\operatorname{csch}^2(\frac{3}{2}H_\infty t)$ and the limit $\operatorname{csch}^2 z\,e^{2z}\to4$ being checked exactly.

5. The lightest mode: dark, collisionless, and stable or not (section 24.3, ESTIMATE)

Every coupling of the lightest mode is diluted by the volume of the torus. With couplings of natural size at the eight-dimensional Planck scale — $\lambda_P=c_P/M_8^5$, $y=c_y/M_8^2$, $G_P=c_V/M_8^6$, $g_r=c_r/M_8^6$ (mass dimensions $-5,-2,-6,-6$; c_P,c_y,c_V,c_r dimensionless) — and $V_\perp=M_{\rm Pl}^2/M_8^6$:

$$\lambda_{\rm eff}=\frac{\lambda_P}{V_\perp}=\frac{c_PM_8}{M_{\rm Pl}^2},\qquad y_{\rm eff}=\frac{y}{\sqrt{V_\perp}}=\frac{c_yM_8}{M_{\rm Pl}},\qquad G_{\rm eff}=\frac{G_P}{V_\perp}=\frac{c_V}{M_{\rm Pl}^2},\qquad G_r=\frac{g_r}{V_\perp}=\frac{c_r}{M_{\rm Pl}^2},$$

numerically $\lambda_{\rm eff}=1.687\times10^{-33}\,c_P$ GeV^{-1}, $y_{\rm eff}=4.107\times10^{-15}\,c_y$, $G_r=1.687\times10^{-37}\,c_r$ GeV^{-2}.

Dark. The operator $\lambda_{\rm eff}\Phi^\dagger\Phi\bar\chi\chi$ contains $\lambda_{\rm eff}v\,h\bar\chi\chi$ ($\Phi^\dagger\Phi=(v+h)^2/2$); the nucleon couples to the Higgs boson with f_Nm_N/v ($f_N\approx0.30$, $m_N=0.9383$ GeV). Exchanging h gives the contact interaction $G_{\chi N}\bar\chi\chi\bar NN$ with $G_{\chi N}=\lambda_{\rm eff}f_Nm_N/m_h^2$, and for a Dirac fermion

$$\sigma_{\rm SI}=\frac{G_{\chi N}^2\mu^2}{\pi}=\frac{\lambda_{\rm eff}^2f_N^2m_N^2\mu^2}{\pi m_h^4},\qquad\mu=\frac{mm_N}{m+m_N};$$

at $m=40$ GeV and $c_P=1$: $\sigma_{\rm SI}=9.56\times10^{-104}$ cm^2, some 56 orders of magnitude below the LZ limit.

Collisionless. A contact self-interaction of strength G_r gives $\sigma\approx G_r^2m^2/(4\pi)$: $\sigma/m=4.94\times10^{-79}$ cm^2/g at $m=1$ GeV ($c_r=1$); the Bullet Cluster allows c_r up to 1.19×10^{39} at $m=1$ GeV.

Stable or not. The Higgs and vector portals are bilinear in χ and cannot make it decay; without the Yukawa portal the lightest mode is absolutely stable. Through the Yukawa portal it mixes with the neutrinos: after symmetry breaking the portal is a Dirac mass $\mu_P=y_{\rm eff}v/\sqrt{2}$ between ν_L and χ ($\mu_P=7.150\times10^{-13}\,c_y$ GeV), the mixing angle is $\theta=\mu_P/m$, and χ decays like a sterile neutrino (rates of a Majorana fermion; a Dirac fermion decays half as fast, immaterial here):

$$\Gamma(\chi\to\nu\gamma)=\frac{9\,\alpha\,G_F^2\,m^5\sin^2 2\theta}{1024\,\pi^4},\qquad\Gamma(\chi\to3\nu)=\frac{G_F^2m^5\theta^2}{96\pi^3},\qquad\sin^2 2\theta\approx4\theta^2,\qquad\frac{\Gamma(\chi\to\nu\gamma)}{\Gamma(\chi\to3\nu)}=\frac{27\alpha}{8\pi}.$$

Since $\theta^2m^5=\mu_P^2m^3$, $\tau(\chi\to3\nu)=K/(c_y^2m^3)$ with $K=2.817\times10^{13}$ s GeV^3. Lifetimes at $c_y=1$:

m	$\tau(\chi\to\nu\gamma)$ (s)	$\tau(\chi\to3\nu)$ (s)
1 GeV	3.59×10^{15}	2.82×10^{13}

m	$\tau(\chi\to\nu\gamma)$ **(s)**	$\tau(\chi\to 3\nu)$ **(s)**
10 keV	3.59×10^{30}	2.82×10^{28}

The lifetime exceeds the age of the universe ($t_0 = 13.79$ Gyr) for $m < m_{\max} = (K/(c_y^2 t_0))^{1/3} = 0.04015$ GeV$\times c_y^{-2/3}$. This is the minimum requirement; searches for the X-ray line of $\chi\to\nu\gamma$ demand far longer lifetimes and only strengthen the conclusions of section 6.

6. The abundance (section 24.4, ESTIMATE; the outcome is OPEN)

6.1 From the solutions of regimes I and II: overproduction

The primordial gas. After the capture of the auxiliary space the gas carries the fraction $f = 0.008728$ of the energy and the radion the rest (for the normalization of the primordial gas; f scales with it). Suppose — most favourably — that the radion energy all became brane radiation at the temperature $T_{\rm RH}$. For radiation $\rho_{\rm rad} = \frac{\pi^2}{30} g_* T^4$ and $s = \frac{2\pi^2}{45} g_{*s} T^3$, so $\rho_{\rm rad}/s = \frac{3}{4}T$ ($g_* = g_{*s}$), and the dark matter needs $\rho_{\rm DM}/\rho_{\rm rad} = mYs/(\frac{3}{4}Ts) = \frac{4}{3}mY/T_{\rm RH}$. The gas exceeds it by

$$O = \frac{f}{1-f}\,\frac{T_{\rm RH}}{\frac{4}{3}mY} = 1.510\times10^7\,\frac{T_{\rm RH}}{\rm GeV}, \qquad O(4\text{ MeV}) = 60418,$$

4 MeV being the lowest reheating temperature compatible with nucleosynthesis.

Radion decay. The radion decays into brane fields through $(\phi/(\sqrt{3}M_{\rm Pl}))T^\mu{}_\mu$,

$$\Gamma(\phi\to f\bar f) = \frac{N_c m_f^2 m_r}{24\pi M_{\rm Pl}^2}\Big(1-\frac{4m_f^2}{m_r^2}\Big)^{3/2}, \qquad \Gamma(\phi\to gg) = \frac{b_{\rm QCD}^2\alpha_s^2 m_r^3}{96\pi^3 M_{\rm Pl}^2} \quad (b_{\rm QCD} = 11-\tfrac{2}{3}n_f),$$

and into Kaluza–Klein pairs of dirac16complex (in the Einstein frame a level has the mass $M_E = e^{-2\beta}\sqrt{m^2+\mu_n^2 e^{-2\beta}}$, hence the coupling $g_n = (2M_n + \mu_n^2/M_n)/(\sqrt{12}M_{\rm Pl})$ to ϕ; four brane spinors per level, $m \ll \mu_n$):

$$\Gamma_n = \frac{3\mu_n^2 m_r}{8\pi M_{\rm Pl}^2}\Big(1-\frac{4\mu_n^2}{m_r^2}\Big)^{3/2}, \qquad \Gamma_{\rm bulk} = \sum_n \Gamma_n \approx \frac{3m_r}{8\pi M_{\rm Pl}^2}\,\frac{V_\perp}{8\pi^2}\Big(\frac{m_r}{2}\Big)^6\frac{B(3,\frac{5}{2})}{2},$$

B the Euler beta function. With $m_r = 10$ GeV: $\Gamma_{\rm brane} = 4.57\times10^{-37}$ GeV, $\Gamma_{\rm bulk} = 6.00\times10^{-24}$ GeV — the radion does NOT become brane radiation: a fraction 0.9999999999999238 of its decays produces dirac16complex, and the brane receives only $\Gamma_{\rm brane}/\Gamma_{\rm bulk} = 7.62\times10^{-14}$ of its energy. Without a later dilution the universe after the radion decay is made of dirac16complex — the opposite of the observed ratio.

Both results mean that dirac16complex as dark matter needs the dilution and reheating that the moduli problem of the minimal model already needs (a later period of inflation on the brane, not part of the model), at a reheating temperature below the normalcy temperature, above which the brane plasma loses its energy to bulk gravitons ($\dot\rho/\rho \sim T^7/M_8^6 > H \sim T^2/M_{\rm Pl}$):

$$T_{\rm RH} < T_* = \Big(\frac{M_8^6}{M_{\rm Pl}}\Big)^{1/5} = 13.26\text{ GeV}.$$

6.2 After the dilution: the portals

Freeze-in. A particle χ made in decays $B\to\chi+\dots$ of a particle B (internal states g_B, mass m_B, partial width Γ) in equilibrium obeys $\dot n_\chi + 3Hn_\chi = g_B\Gamma m_B^2 T K_1(m_B/T)/(2\pi^2)$ (as long as n_χ is far below equilibrium). With $H = 1.66\sqrt{g_*}T^2/M_{\rm Pl}^{\rm (unreduced)}$, $s = \frac{2\pi^2}{45}g_{*s}T^3$ and $x = m_B/T$, $Y = n_\chi/s$ integrates to (Hall, Jedamzik, March-Russell and West)

$$Y = \frac{135\,g_B}{8\pi^3\cdot1.66\,g_*^{3/2}}\,\frac{M_{\rm Pl}^{\rm (unreduced)}\,\Gamma}{m_B^2}\,S(x_{\rm RH}), \qquad S(x) = \frac{2}{3\pi}\int_x^\infty t^3K_1(t)\,dt, \qquad \int_0^\infty t^3K_1(t)\,dt = \frac{3\pi}{2},$$

where $x_{\rm RH} = m_B/T_{\rm RH}$ (instantaneous reheating: production only below $T_{\rm RH}$), $M_{\rm Pl}^{(\rm unreduced)} = 1.22091 \times 10^{19}$ GeV, $g_* = g_{*s} = 106.75$ (below the electroweak bosons $g_* = 86.25$, so for $T_{\rm RH} < m_W$ this understates Y by at most $(106.75/86.25)^{3/2} = 1.4$). For Higgs decays ($B = h$, $g_B = 1$) at $T_{\rm RH} = T_*$: $S(m_h/T_*) = 0.007846$. Other production processes add to these yields; the estimates below are lower bounds on what each portal makes — except for the lightest mode alone, where $S = 1$ is used as an upper bound (up to factors of order one).

Yukawa portal, the lightest mode. $\Gamma(h \to \bar\nu\chi) + \Gamma(h \to \nu\bar\chi) = y_{\rm eff}^2 m_h/(16\pi)$ per channel; with N channels (three lepton flavours and the brane spinors, $N \le 12$) $Y = 9.726 \times 10^{-18}\, c_y^2 NS$. With $Y_1 = Y(c_y = 1)$ the observed abundance needs $c_y^2 = mY/(m\,Y_1)$, which falls like $1/m$, while the lifetime allows only $c_y^2 \le K/(m^3 t_0)$, which falls like $1/m^3$. Both hold only for

$$m < m_* = \Big(\frac{Y_1\,K}{mY\,t_0}\Big)^{1/2} = 1.20 \text{ keV } (N = 1), \quad 4.16 \text{ keV } (N = 12),$$

even with $S = 1$ — below the Lyman-α bound. For example ($N = 1$): at $m = 10$ keV the abundance needs $c_y = 2.12 \times 10^6$ while stability allows $c_y < 254419$; at $m = 10$ MeV it needs $c_y = 67045$ and allows $c_y < 8.05$.

Yukawa portal, the Kaluza–Klein tower. Every level couples to the brane with the same strength as the lightest mode (its mode function at $y = 0$ is also $1/\sqrt{V_\perp}$), so Higgs decays make the whole tower below m_h, $\Gamma(h \to \bar\nu\chi_n) \propto (1 - \mu_n^2/m_h^2)^2$. With $dN = V_\perp \mu^3 d\mu/(8\pi^2)$ and $\int_0^1 2\pi^2 u^3 (1-u^2)^2 du = \pi^2/12$, $\int_0^1 2\pi^2 u^4 (1-u^2)^2 du = 16\pi^2/315$:

$$\sum_n \Big(1 - \frac{\mu_n^2}{m_h^2}\Big)^2 = \frac{V_\perp m_h^4}{192\pi^2} = 7.688 \times 10^{17}, \qquad \sum_n \mu_n \Big(1 - \frac{\mu_n^2}{m_h^2}\Big)^2 = \frac{V_\perp m_h^5}{315\pi^2} = 5.867 \times 10^{19} \text{ GeV},$$

$$Y_{\rm tower} = 7.478\, c_y^2 NS, \qquad \frac{\rho_{\rm tower}}{s} = 570.6\, c_y^2 NS \text{ GeV} \qquad (\text{mean mass } \tfrac{192}{315} m_h = 76.31 \text{ GeV}).$$

(The continuum sums are the limits of the lattice sums with $r_4(k)$: for a radius $R = 100$, $\sum_k r_4(k)(1 - k/R^2)^2$ and $\sum_k r_4(k)\sqrt{k}/R\,(1 - k/R^2)^2$ equal $\pi^2 R^4/12$ and $16\pi^2 R^4/315$ to 10^{-8}.) At $T_{\rm RH} = T_*$ the tower carries more energy than all the dark matter ($\rho_{\rm tower}/s > mY$) unless

$$c_y < 9.88 \times 10^{-6} \qquad (N = 1,\ T_{\rm RH} = T_*),$$

and through the same mixing its states decay into neutrinos and photons long after nucleosynthesis: $\tau(\chi_n \to 3\nu) = 2.82 \times 10^{10}\ \text{s}/c_y^2$ at $\mu_n = 10$ GeV (other channels shorten it by about ten). The Yukawa portal must therefore be very weak or absent; the precise bound from nucleosynthesis and the CMB is not derived here (OPEN). With $c_y < 10^{-5}$ the lightest mode outlives the universe for $m < 86.5$ GeV, and its freeze-in yield is negligible.

Higgs portal. The lightest mode alone gets $Y = 1.99 \times 10^{-49}\, c_P^2 S$ from $\Gamma(h \to \chi\bar\chi) = (\lambda_{\rm eff} v)^2 m_h/(8\pi)$. But the brane breaks the conservation of Kaluza–Klein momentum: $\bar\psi\psi|_W = \sum_{n,n'} \bar\chi_n \chi_{n'}/V_\perp$, so $\lambda_{\rm eff} v\, h \bar\chi_n \chi_{n'}$ couples EVERY pair of levels. For a scalar coupling to two fermions of masses $\mu_1 = u_1 m_h$, $\mu_2 = u_2 m_h$,

$$\begin{aligned}
\Gamma(h \to \chi_n \bar\chi_{n'}) &= \frac{(\lambda_{\rm eff} v)^2 m_h}{8\pi} \big(1 - (u_1 + u_2)^2\big) \lambda^{1/2}(1, u_1^2, u_2^2) \\
&= \frac{(\lambda_{\rm eff} v)^2 m_h}{8\pi} \big(1 - (u_1 + u_2)^2\big)^{3/2} \big(1 - (u_1 - u_2)^2\big)^{1/2}, \\
\lambda(1, x_1, x_2) &= (1 - x_1 - x_2)^2 - 4x_1 x_2 = \big(1 - (\sqrt{x_1} + \sqrt{x_2})^2\big)\big(1 - (\sqrt{x_1} - \sqrt{x_2})^2\big).
\end{aligned}$$

Summed over all pairs with $dN\, dN'$ the sum is $(V_\perp m_h^4/(8\pi^2))^2 = 3.40 \times 10^{38}$ times

$$\begin{aligned}
I_{hp} &= \iint_{u_1 + u_2 < 1} u_1^3 u_2^3 \big(1 - (u_1 + u_2)^2\big)^{3/2} \big(1 - (u_1 - u_2)^2\big)^{1/2} du_1\, du_2 = 9.5385 \times 10^{-5}, \\
I_{hp}^E &= \iint_{u_1 + u_2 < 1} (u_1 + u_2)\, u_1^3 u_2^3 \big(1 - (u_1 + u_2)^2\big)^{3/2} \big(1 - (u_1 - u_2)^2\big)^{1/2} du_1\, du_2 = 7.3808 \times 10^{-5},
\end{aligned}$$

the second with the mass weight $u_1 + u_2$ (mean pair mass $I_{hp}^E/I_{hp} = 0.774\, m_h$). With four brane spinors per level the pairs carry $\rho/s = 2.50 \times 10^{-12}\, c_P^2 S$ GeV: the observed abundance needs

$$c_P = 13.2 \ \ (T_{\rm RH} \gg m_h,\ S \to 1), \qquad c_P = 149 \ \ (T_{\rm RH} = T_*),$$

two orders of magnitude above natural size; for $c_P = 1$ the pairs make 4.5×10^{-5} of the dark matter. Such dark matter would be a population of heavy Kaluza–Klein modes (tens of GeV) whose decays into lighter levels and Standard Model particles through the same portal are not computed here (OPEN).

Vector portal (and any coupling of gravitational strength per pair, $G_{\rm eff} = c_V/M_{\rm Pl}^2$). The rate per pair is of order $G_{\rm eff}^2 T^5$ and the number of pairs lighter than T of order $N(T)^2 \propto V_\perp^2 T^8$, so the production rate divided by $H \sim T^2/M_{\rm Pl}$ is of order (numerical factors omitted; they raise T_V)

$$\frac{\Gamma}{H} \sim c_V^2 \Big(\frac{T}{T_V}\Big)^{11}, \qquad T_V = \Big(\frac{M_8^{12}}{M_{\rm Pl}}\Big)^{1/11} = 492.2 \text{ GeV}, \qquad \Big(\frac{T_*}{T_V}\Big)^{11} = 5.4\times 10^{-18}:$$

negligible. **Dependence on the reheating temperature** (the Boltzmann factor S falls steeply below T_*, and with it what the portals make):

$T_{\rm RH}$ **(GeV)**	$S(m_h/T_{\rm RH})$	**tower: largest** c_y $(N=1)$	**Higgs-portal pairs:** c_P **needed**
5.000	1.25×10^{-8}	0.00783	118265
8.000	4.94×10^{-5}	0.000124	1880
10.00	0.000678	3.36×10^{-5}	508
13.26	0.00785	9.88×10^{-6}	149

6.3 Conclusion on the abundance

With couplings of natural size the minimal model does not produce the observed abundance after the dilution its moduli problem requires: the Yukawa portal must be tiny (else the tower is overproduced), the vector portal is negligible, and the Higgs portal falls short by a factor 22263 (it would need $c_P \approx 149$). Before that dilution the model makes far too much. The origin of the dark-matter abundance is therefore OPEN: it must come from physics outside the minimal model (for example the reheating after the brane inflation that the moduli problem needs, or couplings much larger than natural).

7. dirac16complex and dark energy (section 24.5)

7.1 Matter cannot be dark energy (DERIVED)

Free particles have $p_a = -\partial_a W/(3a^2b^4) \geq 0$, because $W = \omega$ decreases with a. Interacting matter at zero temperature has $p = -\partial E/\partial V$ at fixed particle number N; with $E = \varepsilon V$, $\rho = N/V$:

$$p = \rho\frac{d\varepsilon}{d\rho} - \varepsilon = \rho^2\frac{d}{d\rho}\Big(\frac{\varepsilon}{\rho}\Big) = \mu\rho - \varepsilon, \qquad \mu = \frac{d\varepsilon}{d\rho} = \varepsilon_F \quad \text{(Hugenholtz–Van Hove)}.$$

The pressure is negative only where the energy per particle ε/ρ decreases with density — below the saturation density ρ_0, where ε/ρ has its minimum. There uniform matter has a higher energy than a mixture of droplets at saturation and empty space (both of pressure zero: energy $\rho\,\varepsilon_0/\rho_0 < \varepsilon$), and is metastable or, in the spinodal region $dp/d\rho < 0$, unstable. The uniform Kohn–Sham matter of dirac16complex (section 8 of 7+1.nb: seven space dimensions, Hartree approximation, units $m = 1$, scalar and vector couplings $g_S = -10^5$, $g_V = 10^4$) is

$$\rho = \frac{k_F^7}{105\pi^4}, \qquad c_7 = \frac{1}{15\pi^4}, \qquad \rho_S = c_7\int_0^{k_F}\frac{k^6 m^*}{\sqrt{k^2+m^{*2}}}dk, \qquad m^* = m + g_S\rho_S,$$
$$\varepsilon = c_7\int_0^{k_F} k^6\sqrt{k^2+m^{*2}}\,dk - \frac{(m^*-m)^2}{2g_S} + \frac{g_V}{2}\rho^2, \qquad \varepsilon_F = g_V\rho + \sqrt{k_F^2+m^{*2}},$$

saturating at $k_{F0} = 0.75663615$. Its equation of state $w = p/\varepsilon$ (with $p = \varepsilon_F\rho - \varepsilon$):

k_F	p **(units** m^8**)**	$w = p/\varepsilon$
0.100	1.082×10^{-14}	0.001102
0.300	1.877×10^{-10}	0.008494
0.500	-4.308×10^{-10}	-0.0005296
0.600	-1.028×10^{-7}	-0.03597
0.670	-3.455×10^{-7}	-0.05802

k_F	p **(units** m^8**)**	$w = p/\varepsilon$
0.700	-4.110×10^{-7}	-0.05167
0.750	-1.071×10^{-7}	-0.008446

$p > 0$ at low density, $p < 0$ between about $k_F = 0.5$ and k_{F0} (with a spinodal region), $p = 0$ at saturation; the most negative value is $w_{\min} = -0.05802$ at $k_F = 0.670$ — nowhere near -1, and only in unstable or metastable states.

7.2 A condensate is only a cosmological constant (DERIVED)

A Lorentz-invariant condensate of dirac16complex ($\langle\bar\psi\psi\rangle \neq 0$, which the self-couplings allow) has $T_{\mu\nu} = -\rho_{\rm vac} g_{\mu\nu}$, i.e. $p = -\varepsilon$, $w = -1$ exactly: a contribution to Λ_4, not a dynamical dark energy.

7.3 Vacuum energy on the auxiliary torus (DERIVED, with an ESTIMATE of its size)

The zero-point energy of dirac16complex on the torus (periodic boundary conditions) is a Casimir energy. For a real bosonic field of mass m in eight dimensions, four of them compact with proper size $\ell_\perp$, the finite part of the vacuum energy per unit eight-volume is $-\frac{2}{(2\pi)^4}\sum_{\vec n\neq 0}\left(\frac{m}{\ell_\perp|\vec n|}\right)^4 K_4(m\ell_\perp|\vec n|)$; per unit brane volume ($\times V_\perp$):

$$\rho^{(1)}_{\rm Cas} = -\frac{2}{\ell_\perp^4}\sum_{\vec n\in\mathbb{Z}^4\setminus 0}\left(\frac{m\ell_\perp}{2\pi|\vec n|}\right)^4 K_4(m\ell_\perp|\vec n|) \xrightarrow{m\to 0} -\frac{\Gamma(4)\,Z}{\pi^4\ell_\perp^4}, \qquad K_\nu(z)\to\frac{\Gamma(\nu)}{2}\Big(\frac{2}{z}\Big)^\nu,$$

$$Z = \sum_{\vec n\neq 0}\frac{1}{|\vec n|^8} = \sum_{k\geq 1}\frac{r_4(k)}{k^4} = 8(1-4^{-3})\zeta(4)\zeta(3),$$

the last equality from $\sum_k r_4(k)k^{-s} = 8(1-4^{1-s})\zeta(s)\zeta(s-1)$ (checked against the shell sum to $k = 20000$). A fermion contributes with the opposite sign; dirac16complex has 16 real degrees of freedom on shell, the eight-dimensional graviton $8\cdot 5/2 = 20$. With $\rho_\Lambda = 3H_0^2M_{\rm Pl}^2\Omega_\Lambda = 2.518\times 10^{-47}$ GeV^4:

$$\rho_{\rm Cas}(\text{dirac16complex}, m=0) = +16\,\frac{\Gamma(4)Z}{\pi^4\ell_\perp^4} = 6.762\times 10^{34}\,\rho_\Lambda, \qquad \rho_{\rm Cas}(\text{graviton}) = -8.453\times 10^{34}\,\rho_\Lambda.$$

$m\ell_\perp$	$\rho_{\rm Cas}$**(dirac16complex)**$/\rho_\Lambda$
1	6.057×10^{34}
10	4.200×10^{32}
30	2.469×10^{25}
60	2.301×10^{13}
92.2262	1 (by construction)

It equals ρ_Λ only at $m\ell_\perp = 92.226$, i.e. $m = 59.10$ MeV. Every such term is constant while b is frozen ($w = -1$ up to the radion displacement, 10^{-86}), and all of them add to $\Lambda + \kappa U_0 + \kappa f_W/V_\perp$: the observed Λ_4 needs their sum to cancel to 10^{-35} of each term — the cosmological-constant problem, which dirac16complex does not solve.

7.4 The prediction $w = -1$, and DESI

With the radion frozen the model predicts $w = -1$ exactly. The DESI preference for $w_0 > -1$, $w_a < 0$ (2.8σ to 4.2σ depending on the supernova sample; $w_0 = -0.753 \pm 0.056$, $w_a = -0.84^{+0.23}_{-0.20}$ for DESI + Planck PR4 + ACT DR6 + DES-Y5), if confirmed, would contradict the model whatever dirac16complex is. The radion, the only scalar of the model, cannot be the dark energy either: with $m_r \gg H_0$ it is frozen, and a radion light enough to roll today would change $G_4 \propto e^{-4\beta}$ by order one (ESTIMATE).

8. Verdict (section 24.6)

claim	status	basis
dirac16complex behaves as cold matter ($w = 0$) and can be the Ω_c of the late-universe solution	DERIVED	sections 4.1, 4.3
its effect on the radion is negligible ($\delta\beta = 2.7 \times 10^{-87}$)	DERIVED	section 4.3
as dark matter it needs the stabilizing potential (otherwise decompactification)	DERIVED	section 4.2
dark: $\sigma_{\rm SI} = 9.6 \times 10^{-104}$ cm^2; collisionless: $\sigma/m = 4.9 \times 10^{-79}$ cm^2/g	ESTIMATE	section 5
stable over the age of the universe if $m < 40.2$ MeV$\times c_y^{-2/3}$ (absolutely stable without the Yukawa portal)	ESTIMATE	section 5
mass window: $m > 5.7$ keV (Lyman-α), > 0.077 keV (Pauli, $g = 16$)	data	section 3
primordial gas and radion decay overproduce dirac16complex	DERIVED from the solutions; ESTIMATE for O	section 6.1
Yukawa portal: the Kaluza–Klein tower is overproduced unless $c_y < 9.9 \times 10^{-6}$	ESTIMATE	section 6.2
Higgs portal: the observed abundance needs $c_P = 149$ ($T_{\rm RH} = T_*$)	ESTIMATE	section 6.2
observed abundance produced by the minimal model with natural couplings	OPEN (not produced)	section 6.3
dirac16complex matter as dark energy	excluded (DERIVED: $w \geq -0.06$; $p < 0$ only where uniform matter is unstable or metastable)	section 7.1
dirac16complex vacuum energy or condensate as dark energy	not an explanation (DERIVED: a contribution to Λ_4 with $w = -1$)	sections 7.2, 7.3
evolving dark energy (DESI DR2: 2.8σ to 4.2σ)	in tension with the model ($w = -1$ exactly)	section 7.4

9. Validity and limitations

- The DERIVED statements follow from the field equations of the model and the solutions of the three regimes; they hold for any mass m and any couplings.
- The ESTIMATES assume couplings of natural size at M_8 (c_P, c_y, c_V, c_r of order one), $M_8 = 10$ TeV, $m_r = 10$ GeV, instantaneous reheating at $T_{\rm RH}$, production of the portals by Higgs decays only (other processes add), $g_* = 106.75$ in the freeze-in yield, the continuum limit of the Kaluza–Klein sums (exact to 10^{-8} for $R = 100$), and sterile-neutrino decay rates with the Majorana normalization; order-one factors are not tracked.
- The normalization of the primordial gas (section 4.2) is a choice; the overabundance O scales with it.
- OPEN: the production mechanism of the observed abundance; the precise bound on the Yukawa portal from nucleosynthesis and the CMB; the decays of heavy Kaluza–Klein modes made by the Higgs portal.
- The Casimir energy assumes periodic boundary conditions on the torus and no other bulk fields; it is one term of Λ_4 and does not change the conclusion that the model has $w = -1$.

10. Verification

Every statement above is a check of section 24 of 7+1.nb (Wolfram Language 15.0.1) and of the independent Python program tools/ref_dark.py (numpy, scipy, sympy, mpmath), which uses its own quadratures (the Kohn–Sham integrals by numerical quadrature instead of the closed forms of the notebook), its own root finding and its own lattice sums; the capture and decay results of the solutions are taken from the independent program tools/ref_coupled.py, which it runs. The numbers printed by both programs:

quantity	**7+1.nb**	**Python**	**max. relative difference**	**tolerance**
Ω_c, ρ_c/h^2, s_0, mY	0.2641566, 1.053392×10^{-5}, 2891.284, 4.372005×10^{-10}	0.2641566, 1.053392×10^{-5}, 2891.284, 4.372005×10^{-10}	2.4×10^{-16}	1×10^{-12}
M_D	34050.22	34050.22	0	1×10^{-14}
Pauli bound for $g = 16$	0.07729846	0.07729846	0	1×10^{-14}
radion displacement by dark matter	2.730120×10^{-87}	2.730120×10^{-87}	4.9×10^{-16}	1×10^{-12}
gas share after capture (from the coupled solution)	0.008595793, 0.008728032	0.008595793, 0.008728032	1.5×10^{-10}	1×10^{-8}
τH_b without stabilizer (gas-dominated)	0.3999953	0.3999952	1.4×10^{-7}	1×10^{-5}
λ_{eff}, y_{eff}, G_r	1.686561×10^{-33}, 4.106776×10^{-15}, 1.686561×10^{-37}	1.686561×10^{-33}, 4.106776×10^{-15}, 1.686561×10^{-37}	0	1×10^{-14}
σ_{SI}	$9.556077 \times 10^{-104}$	$9.556077 \times 10^{-104}$	1.3×10^{-16}	1×10^{-12}
σ/m; c_r allowed	4.944225×10^{-79}, 1.189871×10^{39}	4.944225×10^{-79}, 1.189871×10^{39}	1.2×10^{-16}	1×10^{-12}
μ_P; lifetimes	7.150044×10^{-13}, 3.593360×10^{15}, 2.817020×10^{13}, 3.593360×10^{30}, 2.817020×10^{28}	7.150044×10^{-13}, 3.593360×10^{15}, 2.817020×10^{13}, 3.593360×10^{30}, 2.817020×10^{28}	2.8×10^{-16}	1×10^{-12}
largest stable m ($c_y = 1$)	0.04015134	0.04015134	3.3×10^{-12}	1×10^{-9}
gas share f, overabundance O	0.008728032, 1.510442×10^{7}, 60417.69	0.008728032, 1.510442×10^{7}, 60417.69	2.2×10^{-12}	1×10^{-8}
radion decays: bulk fraction, brane share	1.0000000, 7.621336×10^{-14}	1.0000000, 7.621336×10^{-14}	3.3×10^{-16}	1×10^{-9}
T_*, $S(m_h/T_*)$	13.26478, 0.007846132	13.26478, 0.007846132	1.1×10^{-15}	1×10^{-9}
zero mode: Y (Yukawa, Higgs portal)	9.726258×10^{-18}, 1.988947×10^{-49}	9.726258×10^{-18}, 1.988947×10^{-49}	3.2×10^{-16}	1×10^{-12}
m_* ($N = 1$, 12)	1.200004×10^{-6}, 4.156935×10^{-6}	1.200004×10^{-6}, 4.156935×10^{-6}	8.2×10^{-16}	1×10^{-9}
c_y needed and allowed	2.120154×10^{6}, 254419.3, 67045.16, 8.045444	2.120154×10^{6}, 254419.3, 67045.16, 8.045444	6.9×10^{-16}	1×10^{-9}
Kaluza–Klein tower (Yukawa portal)	7.688016×10^{17}, 5.866908×10^{19}, 7.477563, 570.6306, 9.881776×10^{-6}	7.688016×10^{17}, 5.866908×10^{19}, 7.477563, 570.6306, 9.881776×10^{-6}	5.1×10^{-16}	1×10^{-9}
Kaluza–Klein lifetime; largest stable m ($c_y = 10^{-5}$)	2.817020×10^{10}, 86.50344	2.817020×10^{10}, 86.50344	6.6×10^{-16}	1×10^{-9}
Higgs-portal pairs	9.538467×10^{-5}, 7.380787×10^{-5}, 3.404482×10^{38}, 2.502887×10^{-12}, 13.21660 …	9.538467×10^{-5}, 7.380787×10^{-5}, 3.404482×10^{38}, 2.502887×10^{-12}, 13.21660 …	7.5×10^{-14}	1×10^{-8}
Higgs-portal pairs: fraction of the dark matter	4.491756×10^{-5}	4.491756×10^{-5}	7.5×10^{-14}	1×10^{-8}
vector portal: T_V, $(T_*/T_V)^{11}$	492.1942, 5.447550×10^{-18}	492.1942, 5.447550×10^{-18}	0	1×10^{-12}
dependence on T_{RH}	5.000000, 1.248895×10^{-8}, 0.007832489, 118265.1, 8.000000 …	5.000000, 1.248895×10^{-8}, 0.007832489, 118265.1, 8.000000 …	3.8×10^{-14}	1×10^{-8}
Kohn–Sham matter: $w_{\min}$, its k_F, k_{F0}	-0.05801599, 0.6700000, 0.7566362	-0.05801599, 0.6700000, 0.7566362	0	1×10^{-9}

quantity	7+1.nb	Python	max. relative difference	tolerance
Kohn–Sham matter: (k_F, p, w) table	0.1000000, 1.081506×10^{-14}, 0.001101883, 0.3000000, 1.876966×10^{-10} …	0.1000000, 1.081506×10^{-14}, 0.001101883, 0.3000000, 1.876966×10^{-10} …	0	1×10^{-9}
ρ_Λ	2.518266×10^{-47}	2.518266×10^{-47}	5.8×10^{-16}	1×10^{-12}
Casimir energies (massless)	6.762461×10^{34}, -8.453076×10^{34}	6.762461×10^{34}, -8.453076×10^{34}	2.2×10^{-16}	1×10^{-12}
$m\ell_\perp$ and m where the Casimir energy equals ρ_Λ	92.22625, 0.05910239	92.22625, 0.05910239	7.7×10^{-16}	1×10^{-9}
Casimir energy at $m\ell_\perp = 1, 10, 30, 60$	6.056895×10^{34}, 4.199620×10^{32}, 2.469358×10^{25}, 2.300676×10^{13}	6.056895×10^{34}, 4.199620×10^{32}, 2.469358×10^{25}, 2.300676×10^{13}	1.7×10^{-16}	1×10^{-9}

The largest relative difference is 1.4×10^{-7}. The checks of section 24 of the notebook:

#	check	result	section
1	required abundance: m Y = Omega_c h^2 (rho_c/h^2)/s0 = 4.37 10^-10 GeV with rho_c/h^2 = 1.053 10^-5 GeV/cm^3 and s0 = 2891 cm^-3 (to 10^-3)	True	24.1
2	collider bound: with M_Pl^2 = R^n M_D^(n+2) and V_perp = (2 Pi R)^4 = M_Pl^2/M8^6, M_D = (2 Pi)^(2/3) M8 = 34 TeV, above the strongest bound 11.2 TeV	True	24.1
3	a massive dirac16complex particle: eps V == omega, pa V == k_a^2/(3 a^2 omega), pb V == k_b^2/(4 b^2 omega), omega = Sqrt[m^2 + k_a^2/a^2 + k_b^2/b^2]	True	24.2
4	the lightest mode (k_b = 0) is cold matter: pa/eps == k_a^2/(3 (m^2 a^2 + k_a^2)) -> 0, pb == 0, and eps ~ a^-3 at fixed b for k_a = 0	True	24.2
5	dark matter displaces the radion today by H0^2 Omega_c/(2 m_r^2) < 10^-80 (m_r = 10 GeV): the four-dimensional Newton constant is constant	True	24.2
6	regime III: Omega_m a^-3/Omega_Lambda == Csch[3 H t/2]^2 for a = (Omega_m/Omega_Lambda)^(1/3) Sinh[3 H t/2]^(2/3), and -> 4 Exp[-3 H t]	True	24.2
7	couplings of the lightest mode: with V_perp = M_Pl^2/M8^6, lambda_P/V_perp == c_P M8/M_Pl^2, y/Sqrt[V_perp] == c_y M8/M_Pl, G_P/V_perp == c_V/M_Pl^2, g_r/V_perp == c_r/M_Pl^2	True	24.3
8	natural couplings make the lightest mode dark and collisionless: sigma_SI(c_P = 1, m = 40 GeV) < 10^-50 x (LZ limit 2.2 10^-48 cm^2), sigma/m(c_r = 1, m = 1 GeV) < 10^-70 cm^2/g (Bullet Cluster: 0.7 cm^2/g)	True	24.3
9	stability through the Yukawa portal: for c_y = 1 the lifetime chi -> 3 nu equals the age of the universe at m = 40 MeV; it scales as 1/(c_y^2 m^3), and Gamma(nu gamma)/Gamma(3 nu) == 27 alpha/(8 Pi)	True	24.3
10	primordial gas: overabundance O = (f/(1 - f)) T_RH/((4/3) m Y) = 1.5 10^7 T_RH/GeV (6 10^4 at T_RH = 4 MeV)	True	24.4
11	the radion puts more than 1 - 10^-12 of its decays into the bulk (Kaluza-Klein pairs of dirac16complex)	True	24.4
12	Boltzmann factor of production below T_RH: Int_0^Infinity t^3 K_1(t) dt == 3 Pi/2 (exact), S(0) = 1, S(m_h/T*) < 10^-2 with T* = (M8^6/M_Pl)^(1/5) = 13.3 GeV	True	24.4
13	zero mode through the Yukawa portal: the abundance (~ c_y^2 m) and the stability (lifetime ~ 1/(c_y^2 m^3) > age) are compatible only for m < m*, and m* < 5.7 keV (Lyman-alpha) for N = 1 and N = 12 channels even with S = 1	True	24.4
14	Kaluza-Klein tower: Sum_n (1 - mu_n^2/m_h^2)^2 -> V_perp m_h^4/(192 Pi^2) and Sum_n mu_n (1 - mu_n^2/m_h^2)^2 -> V_perp m_h^5/(315 Pi^2) (d^4n = 2 Pi^2 |n|^3 d|n|: exact integrals); the lattice sums with r_4(k) for R = 100 equal Pi^2 R^4/12 and 16 Pi^2 R^4/315 to 10^-8	True	24.4
15	Kaluza-Klein tower through the Yukawa portal: 7.7 10^17 states below m_h with mean mass (192/315) m_h; at T_RH = T* it carries more energy than all the dark matter unless c_y < 2 10^-5 (N = 1)	True	24.4
16	Higgs portal: the zero mode gets Y < 10^-40 (c_P = 1); the pairs of Kaluza-Klein levels (lambda(1, u1^2, u2^2) == (1 - (u1 + u2)^2)(1 - (u1 - u2)^2)) give the observed abundance only for c_P between 10 and 20 (T_RH >> m_h, S -> 1) or between 100 and 200 (T_RH = T*)	True	24.4
17	vector portal: gravitational strength per pair; the production rate of all pairs lighter than T divided by H ~ (T/T_V)^11 with T_V = (M8^12/M_Pl)^(1/11) = 490 GeV is below 10^-17 at T*	True	24.4
18	Kohn-Sham matter of section 8: p > 0 at low density, p < 0 below saturation, p = 0 at kF0, a spinodal region (dp/dkF < 0) in between, and w = p/eps >= -0.06 everywhere	True	24.5

#	check	result	section
19	Epstein sum over Z^4 without 0: Sum |n|^-8 == 8 (1 - 4^-3) zeta(4) zeta(3), compared with the shell sum Sum r_4(k)/k^4 to k = 20000 (tail < 3 10^-8)	True	24.5
20	Casimir energy of dirac16complex on the auxiliary torus: massless 16 Gamma(4) Z/(Pi^4 l^4) > 10^34 rho_Lambda, the massive sum -> the massless value for m l -> 0 (to 10^-4), and equals rho_Lambda at m l = 92.2	True	24.5

The Python program's boolean checks:

#	check	result
1	required abundance m Y = 4.37e-10 GeV (rho_c/h^2 = 1.053e-5 GeV/cm^3, s0 = 2891/cm^3, to 1e-3)	True
2	M_Pl^2 = R^4 M_D^6 with V = (2 pi R)^4 = M_Pl^2/M8^6 gives M_D = (2 pi)^(2/3) M8 = 34 TeV > 11.2 TeV (strongest collider bound)	True
3	massive particle: eps V = omega, pa V = k_a^2/(3 a^2 omega), pb V = k_b^2/(4 b^2 omega)	True
4	lightest mode (k_b = 0): pa/eps = k_a^2/(3 (m^2 a^2 + k_a^2)), pb = 0, eps ~ a^-3 at fixed b for k_a = 0	True
5	dark matter leaves G4 unchanged: displacement < 1e-80	True
6	regime III: Omega_m a^-3/Omega_L = csch^2(3 H t/2) for a = (Omega_m/Omega_L)^(1/3) sinh^(2/3)(3 H t/2), -> 4 exp(-3 H t)	True
7	zero-mode couplings: lambda_P/V = c_P M8/M_Pl^2, y/sqrt(V) = c_y M8/M_Pl, G_P/V = c_V/M_Pl^2, g_r/V = c_r/M_Pl^2	True
8	dark and collisionless: sigma_SI < 1e-50 x 2.2e-48 cm^2, sigma/m < 1e-70 cm^2/g	True
9	stability: lifetime = age at m = 40 MeV (c_y = 1); lifetime ~ 1/(c_y^2 m^3); Gamma(nu gamma)/Gamma(3 nu) = 27 alpha/(8 pi)	True
10	primordial gas overabundance 1.5e7 T_RH/GeV (6e4 at 4 MeV)	True
11	radion decays: more than 1 - 1e-12 into the bulk	True
12	Boltzmann factor: int_0^oo t^3 K_1(t) dt = 3 pi/2 (sympy), S(0) = 1 (mpmath), S(m_h/T*) < 1e-2, 13 < T* < 14 GeV	True
13	zero mode, Yukawa portal: abundance and stability compatible only below m* < 5.7 keV (N = 1, 12)	True
14	Kaluza-Klein tower: int 2 pi^2 u^3 (1-u^2)^2 = pi^2/12, int 2 pi^2 u^4 (1-u^2)^2 = 16 pi^2/315 (sympy); lattice sums agree to 1e-8 (R = 100)	True
15	tower through the Yukawa portal: 7.7e17 states below m_h, mean mass (192/315) m_h, c_y bound < 2e-5 at T*	True
16	Higgs portal: zero mode Y < 1e-40; lambda(1, u1^2, u2^2) = (1 - (u1+u2)^2)(1 - (u1-u2)^2); pairs need 10 < c_P < 20 (S = 1), 100 < c_P < 200 (T*)	True
17	vector portal: 480 < T_V < 500 GeV, (T*/T_V)^11 < 1e-17	True
18	Kohn-Sham matter: p > 0 at low density, p < 0 just below kF0, p = 0 at kF0, a spinodal region, w >= -0.06	True
19	Epstein sum: Sum over Z^4\{0} of |n|^-8 = 8 (1 - 4^-3) zeta(4) zeta(3) = shell sum to k = 20000 (tail < 3e-8)	True
20	Casimir: massless > 1e34 rho_Lambda; massive sum -> massless (m l = 1e-3, to 1e-4); = rho_Lambda at m l = 92.2	True

11. How to reproduce

1. `python3 tools/make_7p1_nb.py` writes 7+1.nb (unevaluated).
2. Evaluate 7+1.nb in a fresh kernel of Wolfram 15.0.1 (Evaluation ▷ Quit Kernel, then Evaluation ▷ Evaluate Notebook; or `Get["D:/Developer/github/7+1/tools/run_eval.wl"]`) and save it; the last section summarizes all checks.
3. `python3 tools/ref_dark.py` repeats the checks of this document independently (exit code 0 if all pass).
4. `python3 tools/make_docs.py` regenerates this document, its symbol dictionary and its provenance record.

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